

Q-1 Define Common Language Runtime (CLR).

-> The Common Language Runtime (CLR) is the core component of the .NET Framework. It acts as a virtual machine that manages the execution of .NET programs. It provides important services such as:

- Memory Management
- Garbage Collection
- Exception Handling
- Security

-> CLR allows different programming languages to run under the same environment.

Base Class Library Support

Thread Support

Com Marshaller

Type Checker

Exception Manager

Security Engine

Debug Engine

JIT
Compiler

Code
Manager

Garbage
Collector

CLR Loader

Q-2 What is the difference between Managed and Unmanaged Code?

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- Managed Code :

Code that runs under the control of CLR. CLR handles memory, security, etc.

Example : Code written in C# or VB.NET

- Unmanaged Code :

Code that runs outside the CLR. You must manage memory and other tasks manually.

Example : Code written in C or C++

Feature	Managed Code	Unmanaged Code
Runs under CLR	Yes	No
Memory Management	Automatic	Manual
Language	C#, VB.NET	C, C++

Q - 3 Explain Common Type System (CTS) in .NET

-> CTS defines how data types are declared, used, and managed in the runtime.

It ensures that objects written in different .NET languages can interact with each other.

Example: An int in C# is the same as Integer in VB.Net

Key Point: It provides type safety and cross-language compatibility.

Q - 4 Write the purpose of web.config file in ASP.NET.

-> The web.config file is the configuration file for an ASP.NET web application.

It stores settings like:

- Database connection strings
- Authentication mode
- Session state configuration
- Custom error pages

Q-5 What is the use of global.asax file?

-> The global.asax file (also called Application File) is used to handle application-level events like:

- Application_Start
- Session_Start
- Application_Error

It is useful for initializing resources or handling errors globally in the application.

Q-6 List any three ASP.NET validation controls with their use.

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1. RequiredFieldValidator - Ensures a field is not left blank
2. RangeValidator - Checks if the input value falls within a specified range
3. RegularExpressionValidator - Validates input using a regular expression

Q-7 Define State Management and list its types.

-> State Management refers to the technique of maintaining user data (state) across multiple requests in a web application.

Types of State Management:

Client-side

- ViewState
- Cookies
- Hidden Fields

Server-side

- Session
- Application
- Database

Q-8 Differentiate between client-side and server-side state management.

Feature	Client-Side	Server-side
Data stored in	User's browser	Server memory
Performance	Faster	Slower, but more secure
Examples	Cookies, ViewState	Session, Application
Data security	Less secure	More secure

Q-9 What is a Master Page in ASP.NET?

-> A Master Page allows you to define a common layout for multiple web pages in ASP.NET.

It contains placeholders (ContentPlaceHolder) that child pages can fill.

Purpose: To provide consistent design and easy maintenance.

Q-10 Define Themes in ASP.NET and their purpose.

-> Themes in ASP.NET are a collection of settings like CSS files, images, and skins that define the look and feel of web controls.

Purpose:

- To maintain consistent UI styling
- To switch UI appearance easily across the site.